Sensitivity analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Ablation studies. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of sensitivity analysis. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **sensitivity analysis**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **sensitivity analysis**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Ablation studies. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **sensitivity analysis** using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Sensitivity analysis case. Separate observations, given facts and assumptions before choosing an explanation.

Sensitivity analysis checks whether a conclusion changes under plausible settings or analysis choices. It should not secretly select the most favourable result.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Sensitivity analysis. Connect each labelled input, deciding step and result.

Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.

Sensitivity analysis — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Retrieval cutoffs1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5.	Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.	Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Sensitivity analysis. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Retrieval cutoffs1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Set the cutoff to exactly 0.6.

Predicted result

Prediction: 0.6 and 0.8 are positive because the rule includes equality.

Why the result follows

Reason: The declared at-least boundary includes the threshold value.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Sensitivity analysis to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Sensitivity analysis	Checking outcomes across declared plausible choices
Robustness range	The conditions over which a conclusion is examined
Post-hoc selection	Choosing a procedure after seeing its scores

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Sensitivity analysis, predict a step, then compare with its explanation.

Retrieval cutoffs1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5.

Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

Worked explanation · Sensitivity analysis

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sensitivity analysis

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

The facts we will use

Supplied facts

Scores are 0.4,0.6,0.8. Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

AI ACADEMY / AI-7/17

Score decisions under inclusive cutoffs

	Score 0.4	Score 0.6	Score 0.8
Cutoff 0.5	No	Yes	Yes
Cutoff 0.7	No	No	Yes
Cutoff 0.6	No	Yes	Yes

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/17

Sensitivity analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Hold scores fixed

Keep the same three supplied scores.



2. Apply cutoff 0.5

0.6 and 0.8 meet the boundary.



3. Apply cutoff 0.7

Only 0.8 meets the new boundary.

CHANGE ONE THING

Set the cutoff to exactly 0.6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Hold scores fixed	Keep the same three supplied scores.
2. Apply cutoff 0.5	0.6 and 0.8 meet the boundary.
3. Apply cutoff 0.7	Only 0.8 meets the new boundary.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

What this example does not establish

Keep the stated limit

Without reference labels, these counts do not identify the best cutoff or establish correctness.

Keep the stated limit

Read the labelled picture: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which

equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

Keep the stated limit

Change one thing: Set the cutoff to exactly 0.6. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

Find the unsupported step in the Sensitivity analysis claim and explain a repair using the supplied evidence.

A stable tested range does not establish stability under every parameter or dataset change.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Sensitivity analysis with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Sensitivity analysis**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Sensitivity analysis**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Run the investigation

> Investigate Sensitivity analysis. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Retrieval cutoffs1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Sensitivity analysis. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **sensitivity analysis evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sensitivity analysis**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Sensitivity analysis case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Sensitivity analysis — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Retrieval cutoffs1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5.	Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.	Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Thresholds .4, .5, .6 give passes8, 9, 7 out of10. Report range and a selection limit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Thresholds .4, .5, .6 give passes8, 9, 7 out of10. Report range and a selection limit.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Sensitivity analysis, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

By the end, I can	How I will show it
Explain the mechanism and terminology of sensitivity analysis	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Sensitivity analysis without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **sensitivity analysis**

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only 0.8 meets the new boundary.

1. Hold scores fixed

2. Apply cutoff 0.7

3. Apply cutoff 0.5

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply cutoff 0.7 Only 0.8 meets the new boundary. **Sensitivity analysis** checks how conclusions change under plausible choices.

Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Set the cutoff to exactly 0.6. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. 0.6 and 0.8 are positive because the rule includes equality.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: 0.6 and 0.8 are positive because the rule includes equality. The declared at-least boundary includes the threshold value.

Repair a missing building block

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

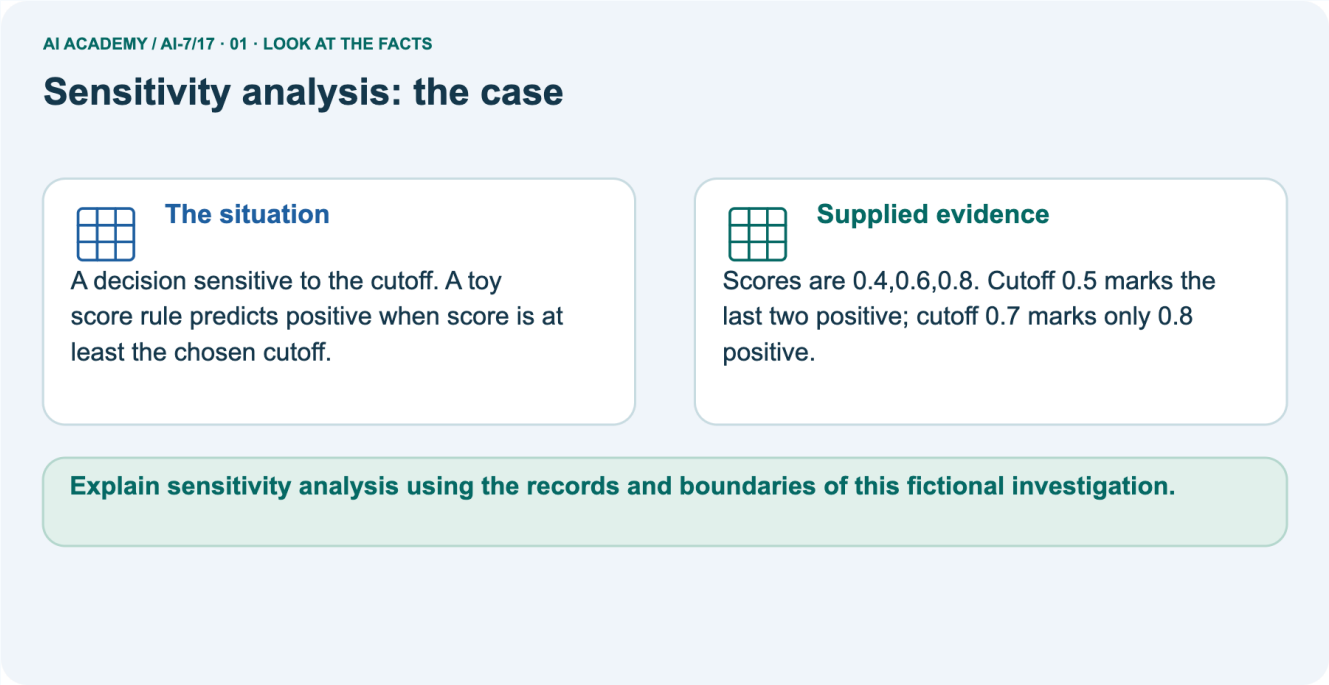
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Sensitivity analysis graphic using its labels, deciding evidence and relevant limitation.

01. Sensitivity analysis: the case



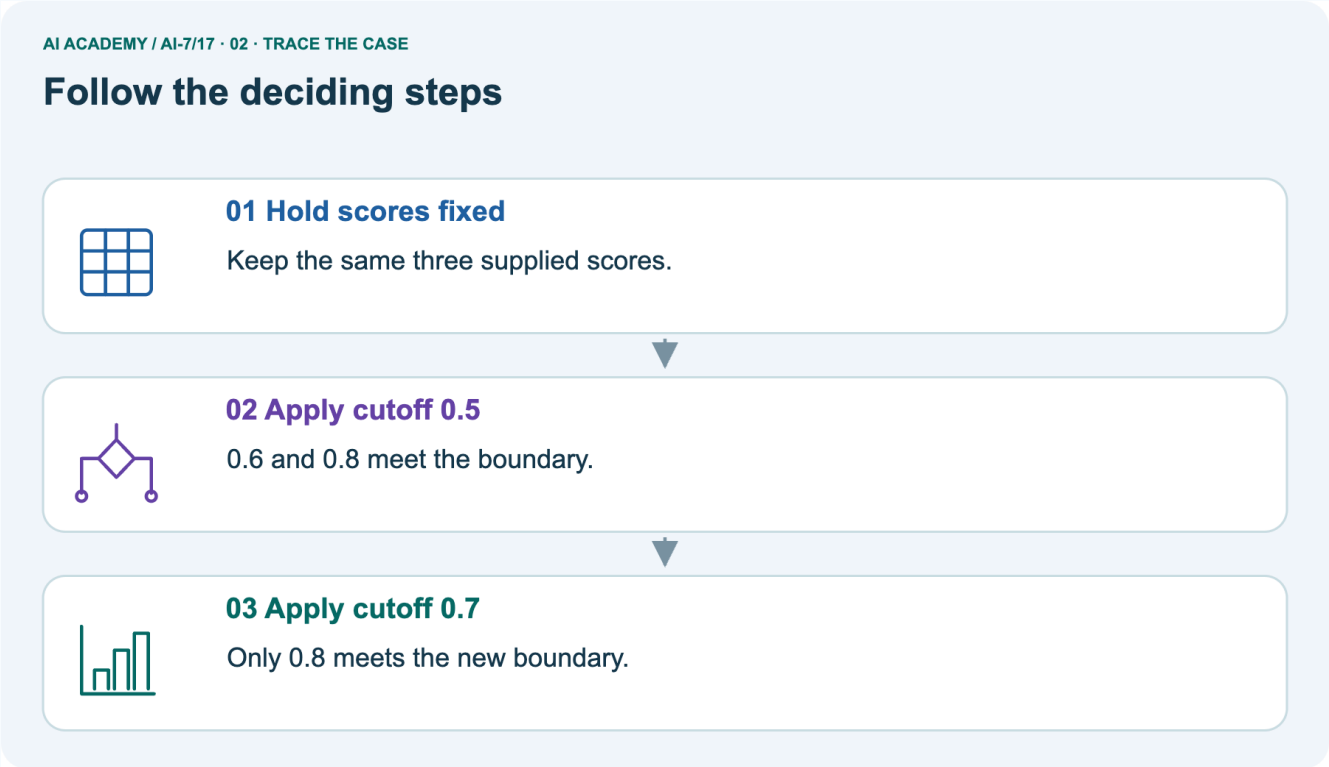
Follow the labelled picture with the matching explanation. Open the vector diagram

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff. Supplied facts: Scores are 0.4,0.6,0.8.

Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

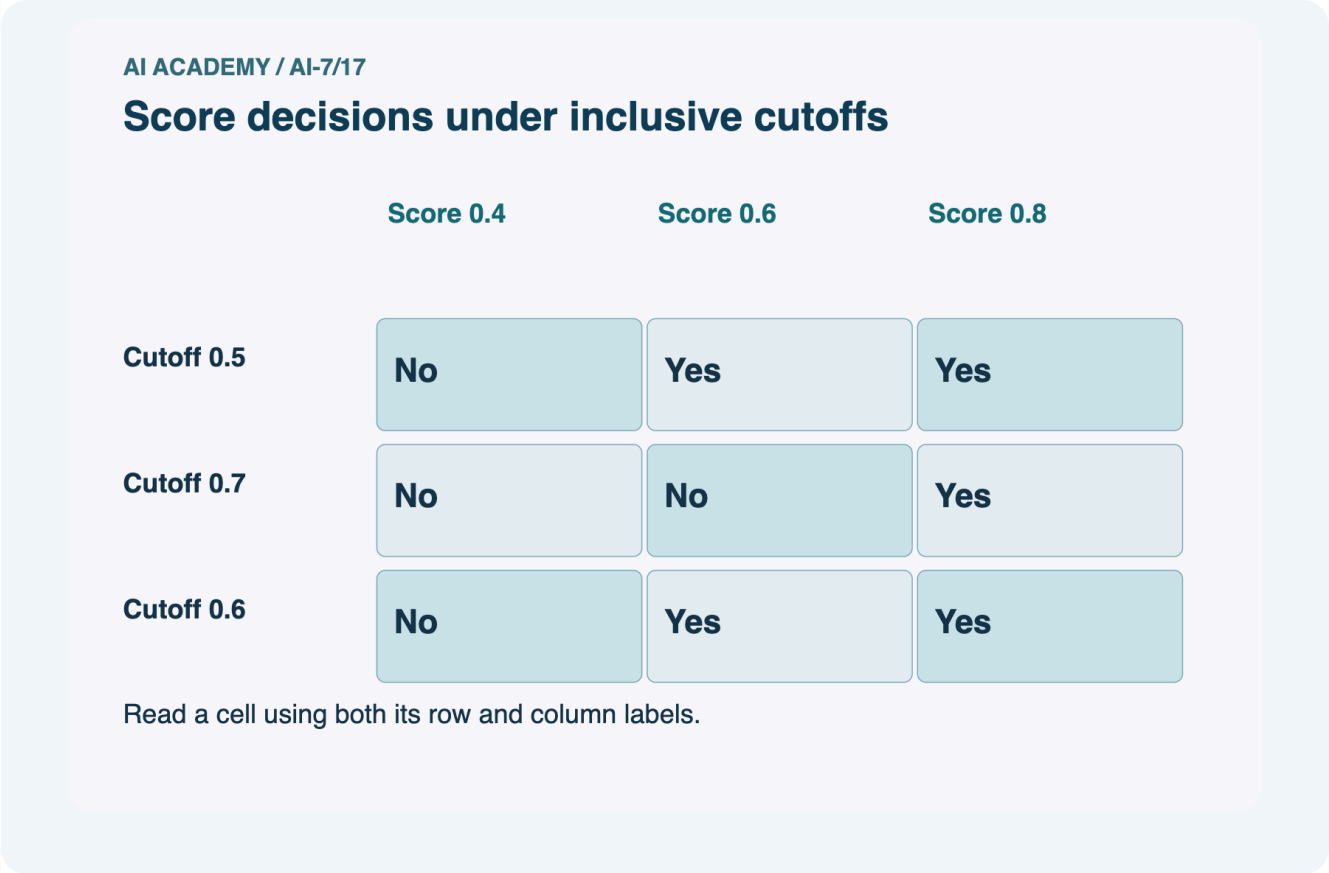


Follow the labelled picture with the matching explanation. Open the vector diagram

Hold scores fixed: Keep the same three supplied scores. → Apply cutoff 0.5: 0.6 and 0.8 meet the boundary. → Apply cutoff 0.7: Only 0.8 meets the new boundary.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Score decisions under inclusive cutoffs

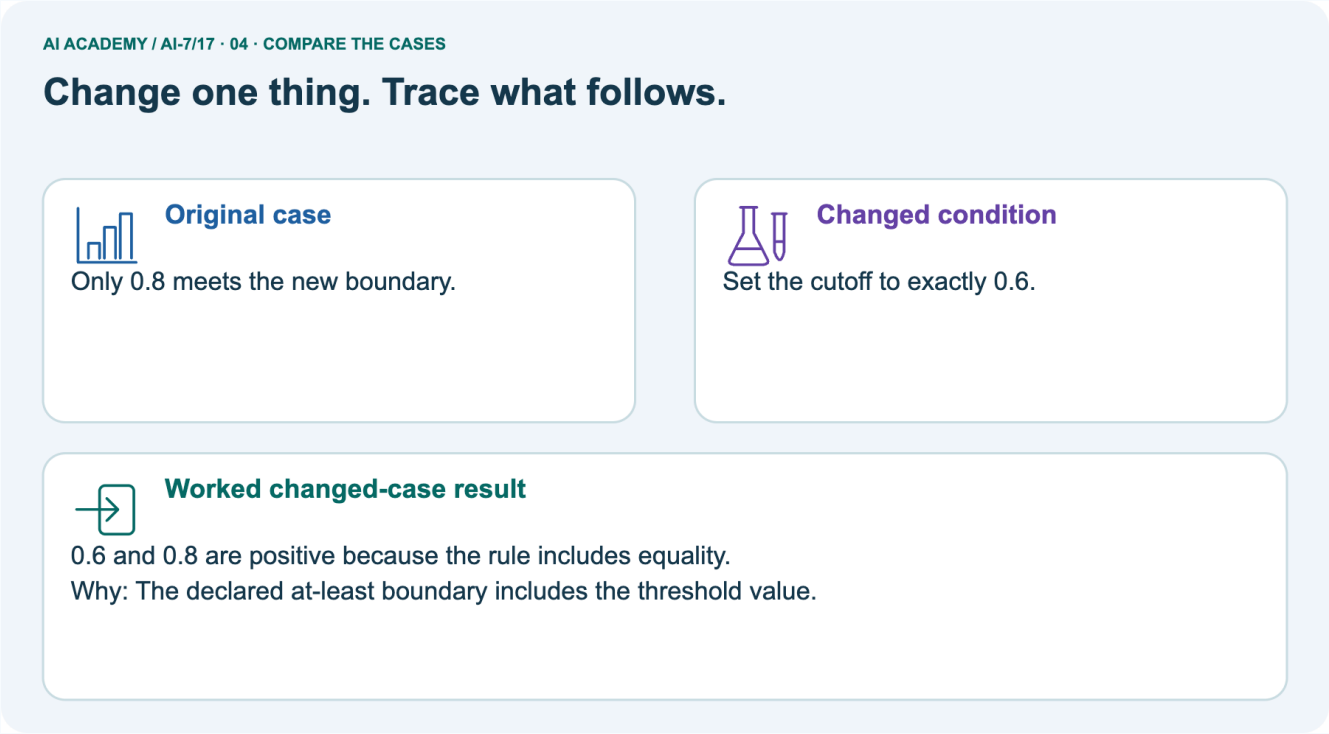


Follow the labelled picture with the matching explanation. Open the vector diagram

Score decisions under inclusive cutoffs. Cutoff 0.5: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes. Cutoff 0.7: Score 0.4 = No, Score 0.6 = No, Score 0.8 = Yes. Cutoff 0.6: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



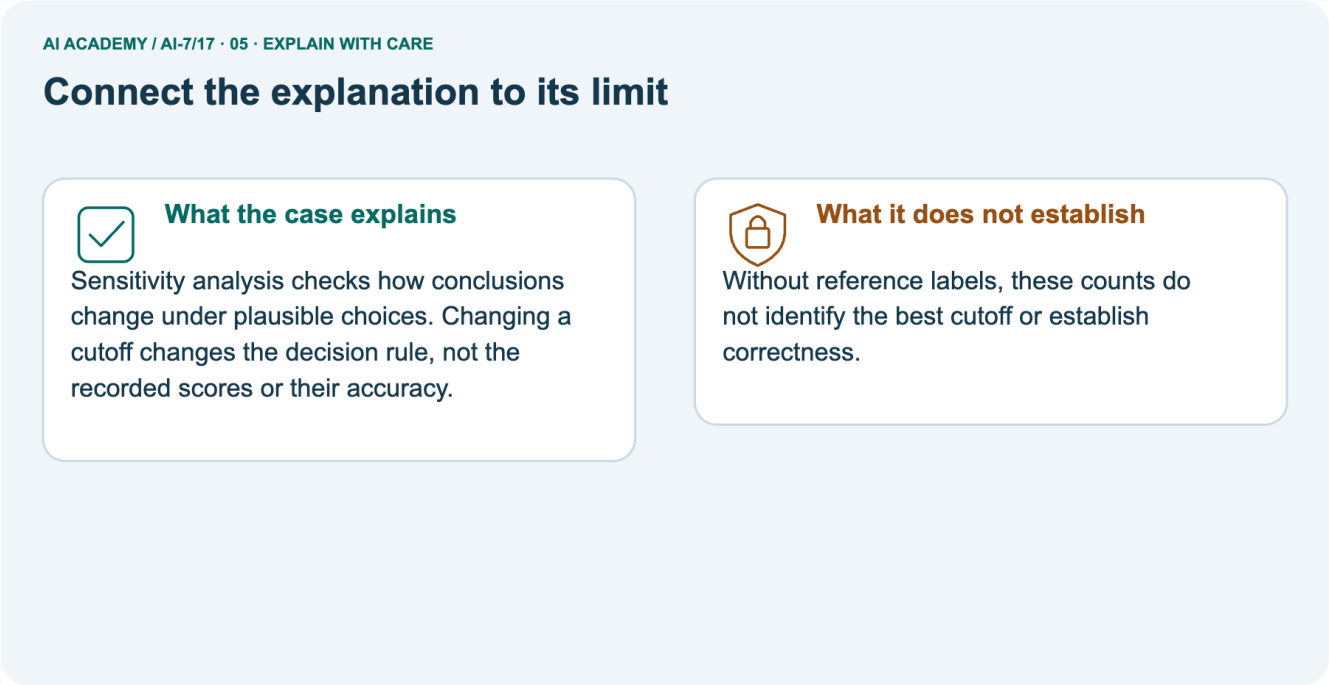
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only 0.8 meets the new boundary. Changed condition: Set the cutoff to exactly 0.6. Expected result: 0.6 and 0.8 are positive because the rule includes equality.

Reason: The declared at-least boundary includes the threshold value.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



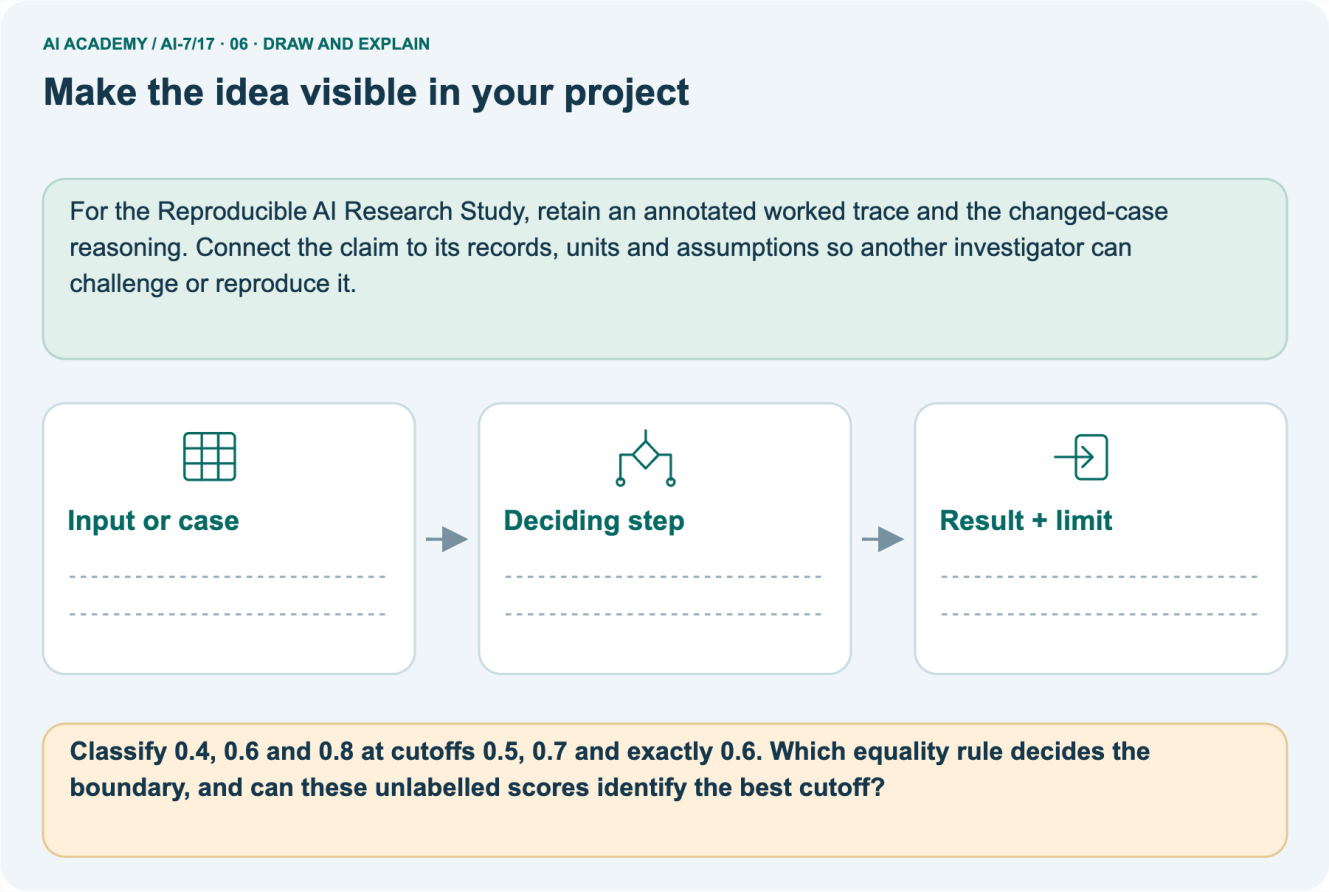
Follow the labelled picture with the matching explanation. Open the vector diagram

Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Limit: Without reference labels, these counts do not identify the best cutoff or establish correctness.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.